

New Construction Report Card

In order to make energy-efficient and comfortable new houses, California builders need to pay more attention to the details.

by **Marc Hoeschele**
and **Rick Chitwood**

Houses are getting more complicated these days, at least in California. Features such as cantilevered floors, interior columns, arches, and soffits complicate the work of the mechanical contractor and the framers, insulators, and drywall contractors, who should be striving for unified thermal, pressure, and moisture (TPM) barriers between conditioned and unconditioned space. Without a carefully installed contiguous TPM, the overall performance of the building will likely be compromised. Furthermore, the pressure to cut costs in the construction industry results in a lack of attention to problems that may be invisible to the homeowner, but critical to the overall house performance, such as low HVAC air flow and sub-par insulation installation.

In recent years, the California Energy Commission (CEC) has worked to improve the quality of residential construction by promoting the use of diagnostic tools and by developing protocols for efficient envelope and HVAC system design and installation. To further this effort, the CEC is sponsoring the Residential Construction Quality Assessment (RCQA) project, which involves detailed diagnostic testing on 60 new homes throughout the state. Geographically, these homes range from San Diego in the south to Mount Shasta in the northeast corner of California, the coldest area in the state.

In phase 1 of the project, we tested 30 homes with tight duct systems. In phase 2, another set of 30 houses was tested. The phase 2 sample included leading edge houses with tight ducts that were previously tested through programs such as PG&E's Comfort Homes, the Sacramento Municipal Utility District (SMUD)'s Advantage Homes, DOE's Energy Star, MASCO's Environments for Living, and DOE's Building America program. We completed phase 1 testing in January 2000, and phase 2



RICK CHITWOOD

testing in the early spring of 2002. Results of the phase 2 testing will be reported in a future article.

A Snapshot of the 30 Homes

Twenty of the phase 1 houses are located in the Central Valley, where much of California's new construction growth is occurring. Eight of the remaining houses are in Southern California, and 2 are in Mount Shasta. Floor areas range from 1,260 ft² to 4,170 ft² with a mean of 2,229 ft². All the houses are slab-on-grade construction; half are single-story and half are two-story houses.

Fourteen of the 30 houses were part of new construction programs that required duct testing. Six were part of a municipal energy efficiency program where a sampling of houses were tested for duct leakage. The remaining ten houses, including two with guaranteed energy bills, were built by builders who made efforts on their own to install tighter-than-normal ducts but did not perform duct testing.

All but two of the houses have 2 x 4 framed wall construction; two have 2 x 6 construction. All 30 homes have studs on 16-inch centers. Wall cavity insulation varies from R-13 to R-19. Nearly half of

the houses have exterior rigid insulation. Three of the houses have cellulose insulation in the wall cavities. Ceiling R-values range from R-19 to R-49, with the 6 R-19 houses located in Southern California.

All 30 houses have split-system air conditioning systems, with all but three of the air handler units located in the attic. The remaining three are located in indoor closets. Two HVAC systems are heat pumps and 30 are gas furnaces with outdoor condensing units (two houses have two HVAC systems). Two-zone systems are installed in five homes. With two exceptions, R-4.2 flex duct is used, mainly in the attic. One house has R-8 ducts fully installed in the conditioned space in hallway dropped ceilings, and the other has uninsulated sheet-metal ducts installed under the building slab.

Detailed HVAC sizing methods, such as ACCA *Manual J* and *D* (or substantially equivalent procedures), were performed by the installing HVAC contractor on eight of the 30 houses.

HVAC Performance

We performed detailed room-by-room load analysis using ACCA *Manual J* software for each of the test houses, if the contractor had not already provided this

analysis. For production home design work, our design loads were based on room-by-room load analysis for the house orientation generating the highest cooling load. We made no effort to aggressively downsize the systems, consistent with how a contractor would perform the sizing calculations.

Cooling system sizing. Cooling system sizing for the test houses averaged 569 ft² per ton and ranged from 440 ft² to 1,010 ft² per ton (the higher the better, as far as tight sizing and intelligent building design are concerned). The two houses with sizing over 1,000 ft² per ton are highly efficient houses with ducts in conditioned space and guaranteed energy bills. Interestingly, the houses located in mild Southern California coastal climates (88°F cooling system design temperature) have lower ft²/ton cooling capacity, on average, than the houses located in the hot Sacramento area (with a 101°F design temperature). This could be an artifact from the small sample size. Some of the Southern California houses are high end, and the builders may have intentionally oversized the systems in an attempt to ensure comfort.

Since *Manual J* analysis was either provided or completed for each of the 30 houses, it was possible to compare installed-system sizes with the sizes based on *Manual J* loads. We determined the required equipment capacity by rounding up to the nearest 1/2-ton increment. Using this procedure, we found that 19 of the 32 installed systems were properly sized, ten were oversized, and three were undersized. We found the average oversizing for the 32 installed systems to be 0.11 tons, or 3% of the average installed capacity.

Manual J assumes duct systems are “substantially tight” and specifies cooling system duct loss of 15% to represent conductive/convective losses for typical R-

4.2 attic ducts. This is a problem for two reasons. First, it doesn't allow the HVAC designer to take credit for a tight duct system (6% leakage) relative to an industry standard duct system (20%–30% leakage). Second, if we presume that *Manual J*

flow is a problem for several reasons. If system airflow is less than the prescribed 350–450 CFM per ton range, refrigerant charging tables become less accurate. Low air flow also increases latent cooling,

which is largely unnecessary and wastes energy in dry climates like California.

On average, the measured HVAC system air flow in the test houses was 349 CFM per ton, or 12.8% below the industry nominal air flow level of 400 CFM per ton. The eight contractor-designed systems had an average measured total air flow within 1% of that specified by *Manual J*, while the 23 nondesigned systems were, on average, 15% below *Manual J* levels. Early on in the phase 1 fieldwork, it became apparent not only that total airflow was typically low (see Figure 1), but also that air flow was poorly distributed within the house, as compared to the *Manual J* room-by-room flow rates. Poor air distribution in a house often results in

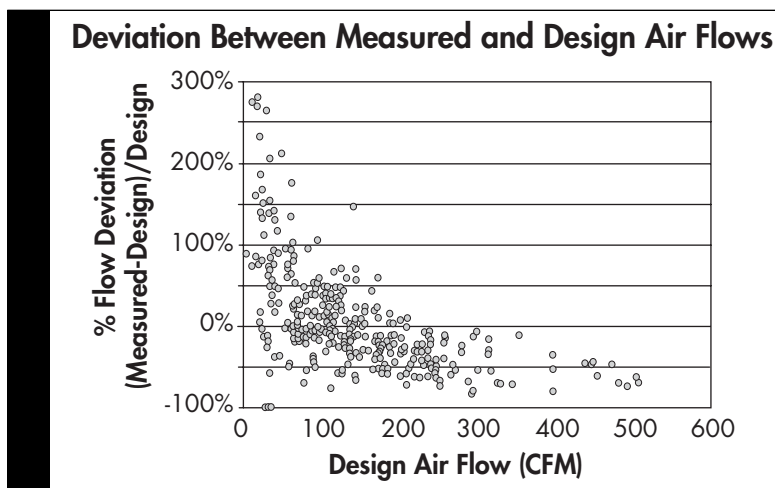


Figure 1. This plot of measured air flow versus design air flow for individual registers shows that registers with low flow requirements consistently get too much air at the expense of the rooms with high air flow needs.

Table 1. SLA Comparison among Identical House Pairs

Pairs	SLA Values	Δ%	Comments
1	1.44 & 1.69	17.4%	2nd value for cellulose wall house
2	3.86 & 4.33	12.2%	1st value for cellulose wall house
3	3.64 & 4.10	12.6%	
4	3.70 & 4.35	17.6%	
5	1.68 & 19.30	14.9%	1st value for cellulose wall house

properly sizes systems, the calculation methodology must include overly conservative assumptions in other areas (envelope heat transfer, solar gains, infiltration) to mask the 20%–30% leakage common to most duct systems. The roughly 14%–24% “slop factor” within *Manual J* increases average system oversizing by roughly 20% for installations with tight duct systems, which amounts to 1/2 ton on systems of 3 tons or more. Under this more realistic scenario, 22 of the 32 systems would be oversized.

Air flow. Although manufacturer-recommended air flow rates for residential split systems are typically 350–450 CFM per nominal ton, actual flow rates were found in a national survey to average 327 CFM per ton (see reference in “For more information” box). Low air

comfort complaints from the occupants, which represent a major source of builder callbacks. For houses that did not have detailed design performed, average measured air flow for rooms with high requirements (for example, the master bedroom suite) was 66% of the design flow, versus 88% (also low) for the designed houses.

Duct leakage. According to CEC requirements, leakage from tight duct systems should not exceed 6% of the system fan flow, in stark contrast to the 20%–30% leakage common to most residential systems. The Duct Blaster fan pressurization device was used to measure duct leakage at each site. Duct leakage for all 32 systems that we tested ranged from 31 to 228 CFM at 25 Pa, with a mean of 106 CFM. As a fraction of nominal fan flow (400 CFM per ton), duct leakage

averaged 7.7%, ranging from 4.1% to 19.6%. Thirteen of the 32 systems satisfied the 6% leakage target. Nineteen of the 32 systems had previously been tested by either the installing HVAC contractor or a third-party tester. On average, contractor-measured leakage was 6 CFM lower than the leakage measured by the RCQA project team.

We did not find that detailed design by the mechanical contractor was a statistically significant indicator of low duct leakage. Interestingly, three of the houses with designed ducts had the highest percentage of leakage. (Usually the designer and installer are different people, which may explain why some of the designed ducts leaked so much.) The Title 24 tight-duct credit requires contractors to test their work as well as to have third-party testing. We did find that RCQA-measured leakage was 39% lower (84 CFM versus 139 CFM) for homes that were previously tested than it was for homes that were not previously tested. If detailed duct design is not a significant factor in preventing duct leakage, contractor testing is significant.

On average, we estimated that 67% of the measured leakage was supply leakage. If the building envelope and the ducts are pressurized to 25 Pa, all measured duct leakage is to outside the building pressure envelope. Duct leakage within the pressure envelope remains in conditioned space, although it may not be directed to where it was meant to go. Duct leakage to the outside, which we measured with the entire house pressurized to 25 Pa with the blower door, was found to average 83 CFM, or 81% of the total duct leakage.

The Building Envelope

In the (generally mild) growth areas of California, less attention may be placed on the insulation installation details and the proper application of air barriers than in other parts of the country. To help evaluate these envelope con-

struction details, we documented envelope performance with an infrared (IR) camera. We also made detailed visual inspections of identical houses that were under construction (with the same floor plan as the completed houses), in order to help us identify the origins of the anomalies we observed.

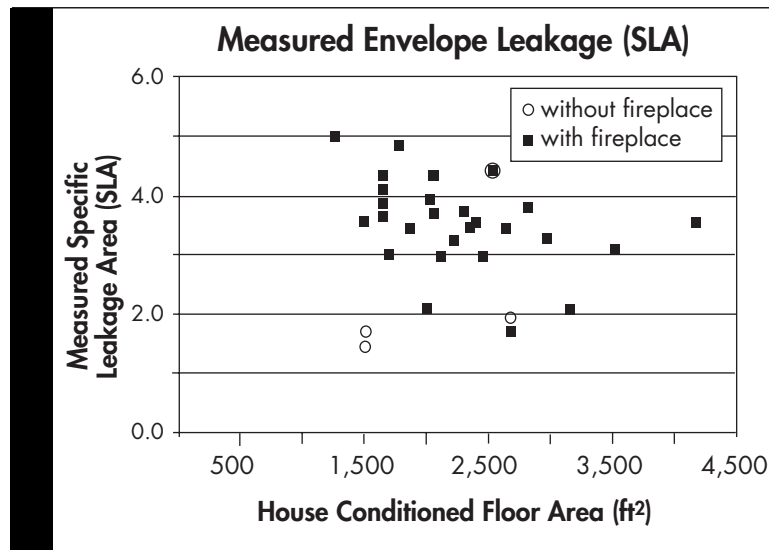


Figure 2. Of the six houses with an SLA of roughly 2 or less, two were built with a special emphasis on tight envelopes, and three were simply constructed with uniform flat ceilings and no soffits or kneewalls.

Envelope leakage. We measured the specific leakage area (SLA) and air changes per hour at 50 Pa (ACH_{50}) for all 30 houses. SLA is essentially the measured blower door leakage normalized by the conditioned floor area of the house. It is calculated by multiplying the measured leakage at 50 Pa (CFM_{50}) by a conversion factor of 3.819, and dividing by the conditioned floor area in ft^2 . The average SLA for all but the two houses with guaranteed energy bills was 3.51 square inches/ ft^2 , which is 20% lower than the average default SLA value assumed in the state energy standards for houses with tight duct systems (4.4 square inches/ ft^2). We completed a final blower door test at each house after sealing all the accessible recessed ceiling lights. By measuring the leak reduction with the recessed can lights sealed, we calculated that recessed lights leaked an average of 11.5 CFM_{50} , or 0.6% of the average measured house leakage per recessed light. The average number of recessed lights per house is nine, ranging from zero (at eight houses) to a maximum of 40.

We completed a statistical analysis to assess the extent to which the presence of specific features would predict the house's SLA. We considered the number of stories, wall cavity R-value, wall insulation type, the presence of exterior-wall foam insulation, ceiling insulation R-value, duct leakage as a percentage of

default fan flow, conditioned floor area, the number of recessed lights, and the presence of a wood-burning fireplace. Given the small sample size, we found that only the fireplace has a statistically significant effect on SLA at the 95% confidence interval. We tested one house with a fireplace after the fireplace dampers and doors were thoroughly taped and sealed. The measured SLA reduction after sealing was 0.72, or 23% of the total house leakage.

The two 1,500 ft^2 houses built with special attention to envelope sealing have an SLA of less than 2 (see Figure 2). Although three of the four houses without fireplaces are among the best sealed, the fourth is one of the leakiest, indicating that there may be other factors significantly affecting house leakage. In addition to the two 1,500 ft^2 houses, we found four additional houses with SLAs of roughly 2. Three of the four houses were simply constructed, with uniform flat ceilings throughout and no soffits or kneewalls. Simple, flat ceiling construction is more conducive to proper air sealing and draft stopping than is typical construction that emphasizes architectural features such as varying ceiling heights, and cantilevered floors. There is no clear explanation for why the fourth tight house (3,150 ft^2 with an SLA of 2) has such low envelope leakage.

We compared calculated SLA values for five identical house plan pairs tested in phase 1 (see Table 1). Interestingly, the SLA difference between the "twins" was never less than 12%, and averaged 15%. This result strongly suggests that random air barrier defects are fairly common. In three cases, one house of each pair had

spray-cellulose-insulated walls. For two of the three house pairs, the cellulose wall had a lower SLA than the batt-insulated wall, although overall there was no statistically significant difference between the batt- and cellulose-insulated houses.

We compared two-story houses to one-story houses with the expectation that the former would have higher leakage due to interior and cantilevered second floors, which are common sources of leakage. For the phase 1 houses, the two-story houses are on average 35% larger than the single-story houses (2,593 ft² for two-story houses versus 1,917 ft² for one-story houses). Interestingly, we found that the two-story houses have 7% lower leakage than the one-story houses (3.61 versus 3.35 SLA). These results may be due to the fact that more envelope leakage can be attributed to the ceiling than to the walls. Also, two-story houses have less surface area/ft² of floor space. Another factor is that larger houses tend to have a lower SLA, since all the houses tend to have features in common (fireplace, vertical plumbing penetrations, vent and flue penetrations) whose impact is reduced as the house size increases.

Based on data collected in phase 1, it appears that—besides fireplaces, a major source of envelope leakage—generic construction defects, such as penetrations through the TPM barrier for plumbing, exhaust fans, and duct chases; leaky interior wall cavities; and complicated floor systems are prime contributors to house leakage.

Insulation inspection. We performed visual and IR camera inspections to assess both wall and ceiling insulation quality. When possible, we made efforts to maximize the value of the IR imaging by heating up the house the day prior to testing and performing IR scans first thing the next day.

We concluded from our visual and IR inspections that the basic task of filling a clear wall cavity (a cavity with no plumbing, minimal wiring, regular stud spacing, and so on) with insulation is performed better than the handling of details that pertain to kneewalls, skylight shafts, cathedral ceilings, draftstopping, and cavities that are unusually sized or that have plumbing and wiring obstructions. These details require more time and attention and are therefore often overlooked during the rapid wall insula-

tion construction phase. The visual observations of the predrywall homes of the same models as the completed homes we tested indicate more extensive defects than were indicated by IR imaging. This is largely due to the underestimating of defects with the IR camera under conditions with small indoor-to-outdoor temperature differences.



Looking at a kneewall from the attic side we see the common problem of the batt falling away from the drywall. Kneewall performance is also hurt because the batts were only run to the ceiling level. If the batts had been installed 10 inches higher, they would act as a dam for the ceiling insulation.

We inspected the ceiling insulation visually. For batt insulation, R-value was determined based on batt thickness and labeling. For blown-in insulation, we recorded insulation depth at a minimum of three attic locations to account for variations in depth. Average depth was compared to the required depth based on the type of blown-in insulation. For the 26 houses where blown-in attic insulation was installed, insulation depth averaged 93% of the required depth. Five houses had less than 80% of the required depth, while eight houses had more than the required depth.

In 1999 the California Institute for Energy Efficiency (CIEE) developed envelope protocols in conjunction with industry experts to improve the quality of insulation installation, air sealing, and window installation. To see how well individual elements of the protocols are being followed in the field, we scored each item in the CIEE envelope protocol by estimating how well each item was completed in the field (see Table 2).

Interior wall cavity pressurization testing. The goal of our building cavity

depressurization studies was to measure cavity pressures at as many accessible locations as possible while depressurizing the house to 50 Pa with respect to outdoors. Ideally, if the cavities are fully within the house pressure envelope, the measured pressures would be 0.

Commonly tested areas include fireplace chases; duct chases; the wall cavity behind the HVAC thermostat; floor systems; under stairs; house architectural features (columns, arches, pillars, and soffits); and other interior wall cavities.

Most of the cavities, with the exception of the wall area behind the thermostat and under the stairs, are closer to outdoor than to indoor conditions. Fireplace and duct chases were, on average, most connected with outdoors. Although these results clearly indicate a stronger thermal connection to outdoors than previously thought, they do not quantify the annual energy impact of the thermal connection. Many factors affect how these cavities interact with conditioned and unconditioned space. These factors include the size of the cavity, the local climate, and pressure imbalances caused by the operation of the HVAC system fan.

A Final Grade?

An integrated “house as a system” approach to production home building is needed to improve the energy performance and comfort of new homes in California. Houses are generally designed and oriented with little consideration given to the local climate. Architectural features such as turrets, soffits, and interior arches are added with little thought as to how they will be insulated and draftstopped in the field. The various subcontractors typically work independently of one another with little knowledge of the impact that their work has on overall house performance. For example, the plumber or electrician may punch gaping holes in draftstopping that the framer has carefully installed. Duct chases, interior columns, and other wall cavities are often open to unconditioned attic space. The result is that, in general, the TPM barrier is not being installed in a continuous manner, resulting in degraded envelope performance.

The HVAC contractor is ultimately left the responsibility of ensuring that

Table 2. Comparison of Field Observations with Envelope Protocols

CIEE Envelope Protocol Item	% Time Properly Implemented
Wall Insulation	
Batts correctly sized to fit snugly at sides and ends	70%–90%
Nonstandard width cavities with insulation cut 1 in wider than cavity	70%–90%
Rim joists insulated to wall R-value	70%–90%
Kneewall batts in contact with drywall	70%–90%
No stuffing of insulation in nonstandard width cavities	40%–60%
Minimal insulation compression	40%–60%
Tub/shower walls insulated	40%–60%
2 x 6 cavities with R-13 instead of R-19	70%–90%
Rim joist insulation cut to fit	10%–30%
Exterior wall channels insulated	10%–30%
Kneewalls/skylights insulated to minimum R-19	10%–30%
Insulation cut to fit around wiring, plumbing, electrical boxes	0%
Skylight shaft batts in contact with drywall	0%
Attic side of kneewall/skylight batts covered by a facing rated to stop attic air intrusion	0%
Ceiling Insulation	
1 in free air space between roof sheathing and insulation at eave/soffit vents	100%
Baffles at eave/soffit vents installed to keep insulation from blocking vents	70%–90%*
Non-IC fixtures boxed and insulated	70%–90%
Insulation covering all IC-rated light fixtures	70%–90%
Blown-in insulation installed uniformly and to required thickness	40%–60%
Draftstops in place over all deep drops and interior wall cavities to stop air movement	10%–30%**
Attic access insulated with rigid or batt insulation	10%–30%
Ceiling batts cut or split to fit around wiring and plumbing	0%
Caulking and Sealing Procedures	
Top plate penetrations sealed	100%
Weatherstripping at exterior doors	100%
Continuous sealing at bottom plate	40%–60%
Weatherstripping at attic access	10%–30%
Sealing around tub and shower drains to the floor	10%–30%

* Baffles are often poorly installed and do not prevent wind from blowing insulation from vents.

** Building-code-compliant draftstops are often in place, but rarely are they airtight.

comfort requirements are met. To do so, he or she needs to plan on some level of envelope defects within each house. Load calculations are fine, but they don't handle the construction anomalies that the contractor must account for (whether or not they exist). The housing industry needs improved sizing tools that accurately account for all the sources of load (and mitigation measures such as tight ducts and radiant barriers). When oversized equipment is installed, it reduces system efficiency. HVAC performance is further degraded by low air flow, poor air distribution, and incorrect refrigerant charge. If comfort complaints occur, increasing the cooling capacity is the most common response.

Despite the deficiencies noted in this study, it is important to point out that there are many well-intentioned builders and contractors committed to building quality homes in California. We found examples of tight duct systems, good HVAC air flow, and tight building envelopes. Putting it all together is the hard part and requires diligence on the part of sub-contractors and field supervisors. Improved training protocols and HVAC design tools, field training of subcontractors, tighter builder bid specifications, increased compensation for high-quality work for subcontractors (whose first priority has become speed), and third party verification are all necessary to improve the performance and comfort of new homes in California.



Marc Hoeschele has worked at Davis Energy Group for 18 years performing a wide range of monitoring and analysis of residential building systems and HVAC technologies. Rick Chitwood is an HVAC and insulation subcontractor who uses a systems approach and performance testing to provide his customers with comfortable and energy-efficient homes.

For more information:

Neme, C., Proctor, J. and Nadel S. 1999. *National Energy Savings Potential From Addressing Residential HVAC Installation Problems*. Report for the U.S. EPA Energy Star program.

Bruce Maeda at the CEC was the contract manager for this project. Tel: (916)654-4077; E-mail: BMaeda@energy.state.ca.us.